CLAIMPROMOTEBENCH: SOURCE-TRACEABLE REGRESSION TESTS FOR MODEL SERVICES UNDER POLICY REVISION

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ABSTRACT

Versioned repository policies can change the required action for an unchanged request. We present CLAIMPROMOTEBENCH, a regression-testing protocol for synthetic, policy-constrained evidence-state transitions. It converts finite policy differences into paired route tests and reports changed-rule and action-transition coverage, a cardinality-minimum structural suite, parser provenance, and failure-inclusive item and pair accuracy. Eight public revision histories anchor the tested fault classes. Two independently authored `AGENTS.md` changes provide model-facing policy sources; their ten author-mapped pairs were fixed before calls to eight hosted routes. The designated first replication yields 132/160 valid decisions, 128/160 correct items, and 52/80 correct pairs. Changed pairs score 23/48 end to end and 23/26 when both decisions are valid; stable controls score 29/32 and 29/29. A matched 15-factor diagnostic gives route-level pair accuracy from 0/15 to 11/15; 280/311 failures over three repetitions coincide with an output limit. Earlier evidence-state studies show why action balance matters: an imbalanced route reaches 258/300 items and 113/150 pairs, while a balanced challenge yields 132/180 and 90/180 items for two routes, with `DEMOTE` recalls of 4/30 and 0/30. The estimand is route conformance on frozen cases. Deployment incidence and downstream utility remain open.

1 INTRODUCTION

Repository context files are versioned inputs to coding agents. One study finds them in 466 open-source projects; another identifies 2,303 files across 1,925 repositories (Mohsenimofidi et al., 2026; Chatlatanagulchai et al., 2025). Controlled evaluations show that agents follow these files and that their effects on task success, cost, and latency vary (Lulla et al., 2026; Gloaguen et al., 2026; Khatri, 2026). A repository can therefore change the policy governing an agent while leaving the user's request unchanged.

Two public histories make the regression concrete. Open Policy Agent added vulnerability-triage guidance to `AGENTS.md`; WooCommerce Services added backward-compatibility guardrails after an interface break (Open Policy Agent, 2026; WooCommerce Services, 2026). Other histories change executable admission policy, verifier semantics, or guardrail adapters (Open Policy Agent Gatekeeper Library, 2020; 2021; Aqua Security Trivy Checks, 2024; Sigstore Cosign, 2022; Microsoft Agent Governance Toolkit, 2026; Agentgateway Project, 2026). In each case, source version belongs in the test fixture because it determines the expected downstream action.

Existing context-file studies vary file presence or construction strategy. Policy-differencing and behavioral-testing methods produce change witnesses, while structured-output benchmarks measure interface completion. The missing test object joins these layers: the same case under two source versions, the prescribed action transition, and the complete model-service route that must deliver it.

CLAIMPROMOTEBENCH builds this object. For executable policies, it enumerates a declared finite context set, retains action-changing contexts, certifies changed-rule and directed-transition coverage, and selects a cardinality-minimum structural suite. For prose policy, it binds immutable source versions to an explicit author mapping. A *route* includes the model identifier, provider, API adapter, prompt, parser, output contract, and execution time. Every scheduled call contributes to end-to-

end item and pair metrics; conditional metrics describe the subset that produced valid decisions. Users remain responsible for policy parsing, grammar synthesis, rule-change discovery, and witness inference.

The resulting object is a version-control analogue of a behavioral regression test. A source diff defines the expected action transition, the paired case tests whether the route follows that transition, and a stable control tests whether the route reacts indiscriminately to source change. This joint object is absent from context-file evaluation, policy differencing, and structured-output testing in isolation.

The evaluation combines three forms of evidence. Eight public histories establish source-level fault classes. Two external `AGENTS.md` revisions test model-facing version sensitivity on eight routes. Synthetic evidence-state instruments isolate 15 policy factors, action imbalance, parser choice, and repeated route stability. The deterministic policy executor is the definitional policy ceiling. Learned routes and a weak Naive Bayes control are empirical measurements.

The contributions are:

1. a benchmark object that binds policy-version pairs to relative rule and transition certificates and a cardinality-minimum structural suite;
2. a route-level scoring protocol that separates raw-strict, registered-parser, contract-action, and exact-decision endpoints while preserving complete denominators; and
3. a source-traceable evaluation over eight upstream histories, external model-facing policy revisions, repeated hosted routes, and action-balanced evidence-state diagnostics.

The finite enumerator uses established semantic-differencing and set-cover machinery. The technical contribution is their integration with version-bound route records, parser provenance, and failure-inclusive endpoints.

2 Related Work

Repository context for coding agents. Empirical studies characterize repository context files as evolving configuration artifacts and measure their presence against task completion, latency, token use, and tool behavior (Mohsenimofidi et al., 2026; Chatlatanagulchai et al., 2025; Lulla et al., 2026; Gloaguen et al., 2026; Khatri, 2026). Their controlled coordinate is usually context-file presence or construction strategy. We hold the route and case fixed, vary a naturally occurring policy version, and score the action transition prescribed by its changed clauses.

Behavioral tests, contrasts, and metamorphic relations. CheckList organizes behavioral tests by capability, and contrast sets probe local decision boundaries through controlled input changes (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing evaluates relations among transformed executions (Chan et al., 1998). The policy-version pair is a contrast whose relation is defined by an executable evaluator or a declared source-to-action mapping.

Conformance, semantic differencing, and regression. Labelled-transition-system and finite-state-machine methods derive conformance tests from explicit behavioral models (Tretmans, 1996; Chow, 1978). Semantic differencing identifies behavioral changes between test-design models (Tzoref-Brill & Maoz, 2017). Margrave computes witnesses for access-control policy disagreement, while Cirg uses change-impact analysis to generate and reduce policy tests (Fisler et al., 2005; Martin & Xie, 2007b). Mutation testing and suite reduction supply established adequacy machinery (Martin & Xie, 2007a; Harrold et al., 1993). Prompt regression measures service evolution under a fixed application task (Ma et al., 2024). ClaimPromoteBench uses the policy version as the controlled coordinate and carries each witness through a model-service contract.

Rule and specification following. RuleArena and SpecEval test rule-governed and specification-following behavior (Zhou et al., 2025; Ahmed et al., 2025). GuideBench includes guideline updates, and TDAD studies specification evolution with hidden tests and semantic mutants (Diao et al., 2025; Rehan, 2026). Policy-to-Tests translates governance prose into executable rules (Datla et al., 2025). Policy-document benchmarks study comprehension, internalization, request compliance, and evolving moderation rules (Bao et al., 2026; Liu et al., 2026a; Cisneros-Velarde, 2026; Mullick et al.,

2023; Ji et al., 2026). COMPASS and augmented policy training study organization-specific policy alignment and guardrail generalization (Choi et al., 2026; Liu et al., 2026b). Our endpoint is the action prescribed by a registered source revision on a fixed case.

Structured output and selective prediction. Grammar-constrained decoding and incremental parsing improve syntactic validity (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench and BFCL evaluate schemas and tool arguments (Geng et al., 2025; Patil et al., 2025). Selective prediction evaluates risk under a learned reject option (Geifman & El-Yaniv, 2017; 2019). Our action set contains an explicit `HOLD` decision; route failures remain terminal outcomes outside that action set.

Research agents. ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, reproducibility, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). Our auxiliary evidence-state suites isolate one downstream policy transition within such workflows.

3 METHOD

3.1 FINITE POLICY-DIFFERENCE ENUMERATION

The benchmark takes deterministic evaluators P_0 and P_1, a finite set $\mathcal{X}$ of complete typed contexts with stable identifiers, registered changed-rule witnesses, and the rule inventory to cover. Let $\pi_P(c)$ denote the prescribed action. The enumerator executes both evaluators and defines

$$\Delta(P_0, P_1; \mathcal{X}) = \{c \in \mathcal{X} : \pi_{P_0}(c) \neq \pi_{P_1}(c)\}.$$

Each $c \in \Delta$ yields a pair $((P_0, c), (P_1, c))$. The context, query, and output contract are identical; only the policy source changes. This construction makes policy-version sensitivity a necessary condition for pair correctness. A deterministic route invariant to the policy returns the same action twice, so at most one member can be correct and pair accuracy is zero.

The enumerator associates every candidate with its registered changed-rule witnesses and directed action edge $(\pi_{P_0}(c), \pi_{P_1}(c))$. For suite T, adequacy is

$$A(T) = \left(\frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where R_Δ is the registered changed-rule inventory and E_{reach} is the action-edge set observed by exhaustive execution over $\mathcal{X}$. Both certificates are relative to the declared context set and registered witnesses. Their validity depends on context completeness and witness correctness.

Selecting the smallest structural-coverage suite with $A(T) = (1, 1)$ is a set-cover instance. We use a standard exact bit-mask dynamic program with $O(n2^{r+k})$ state updates for n candidates, r rules, and k edges. A specialized search selects one context per edge while tracking rule masks; a complete k-pair witness meets the edge lower bound and is globally minimum for the candidate pool. Appendix A gives the proof and optimality certificate.

For versioned prose, we replace the executable oracle with a declared mapping $M_v(c) \in \mathcal{A}$ from a pinned source version and case to a finite action. Source authorship, case authorship, and mapping authorship are recorded separately. A changed pair satisfies $M_0(c) \neq M_1(c)$; a stable control satisfies equality. These pairs test source-version sensitivity under the declared mapping. Rule-coverage and minimum-suite claims apply only where an executable evaluator and registered witnesses define the coverage universe.

3.2 PARSER AND CONTRACT ENDPOINTS

For scheduled item x, let $y(x) = \pi_P(c)$; let $r(x)$ be the action parsed directly from retained raw text; let $p(x)$ be the action recorded by the route-specific registered parser; let $v(x)$ indicate full output-contract validity; and let $d(x)$ be the parsed structured decision. The raw-strict parser accepts exactly one duplicate-free top-level JSON object and performs zero fence or substring extraction, repair, coercion, or non-standard constant handling. Registered parsers retain their historical route-specific

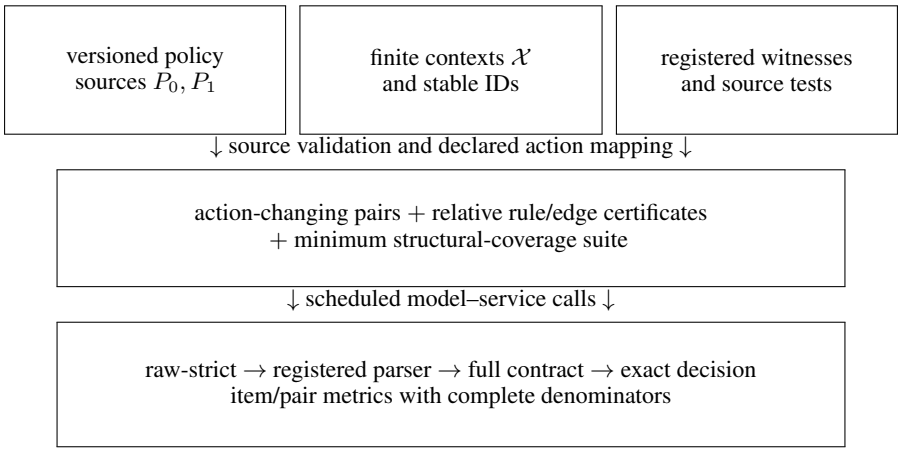


Figure 1: Benchmark and evidence flow. Declared inputs precede enumeration; source validation and suite construction finish before model calls. Each scoring layer corresponds to a distinct interface requirement.

behavior. Their provenance distinguishes direct parsing from deterministic single-object extraction. Every retained response has unique top-level keys. We report:

$$
\begin{aligned}
C_R(x) &= \mathbb{1}[r(x) = y(x)], && \text{raw-strict action,}\\
C_P(x) &= \mathbb{1}[p(x) = y(x)], && \text{registered-parser action,}\\
C_C(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[p(x) = y(x)], && \text{contract action,}\\
C_F(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[d(x) = d^*(x)], && \text{full decision.}
\end{aligned}
$$

The task-specific contract contains five fields for Gatekeeper and six fields for the evidence-state tasks. Pair correctness is the product of the two member indicators for the selected endpoint. End-to-end metrics divide by every scheduled item or pair. Conditional item metrics divide by items for which the selected parser or contract succeeds; conditional pair metrics require both members to succeed. Transport, syntax, schema, and semantic failures are terminal route outcomes. They appear as `N` in confusion matrices, outside the policy action set, and remain incorrect in end-to-end metrics.

4 Evidence and Evaluation Design

4.1 Source-traceable revisions

The model-facing study uses two public `AGENTS.md` histories. Open Policy Agent adds vulnerability-triage guidance, and WooCommerce Services adds backward-compatibility rules after an interface break. We pin each change and parent, verify source and license hashes, and retain the maintainer review record. Ten author-constructed cases yield six action-changing pairs and four stable controls. Upstream maintainers authored the policy text; we authored the scenarios, finite actions, precedence rule, and labels. All were commit-frozen before replacement inference. The study uses author-constructed labels, making inter-annotator agreement inapplicable.

Executable histories supply exact source checks. Open Policy Agent 1.19.1 reproduces two Gatekeeper changes on 12 contexts derived from upstream tests. We map zero violations to `ALLOW` and one or more to `HOLD`. Trivy contributes four regression cases around an inverted predicate; Cosign contributes six verifier cases spanning four interface faults. Every source, parent and change commit, upstream test name, and license is retained.

A fixed query screened 465 public repository seeds and inspected 11 model–policy candidates at source level. Microsoft Agent Governance Toolkit and agentgateway met the declared boundary, licensing, and evidence criteria. We execute the former's parent and fixed Rego sources. For the latter, we inspect the parent source and execute 33 selected tests at the fixed commit. This convenience sample establishes concrete boundary faults; prevalence remains unmeasured.

Table 1: Evidence layers. Structural minima are exact relative to declared candidate sets.

Layer	Source or instrument	Role	Scope
External prose policy	OPA and Woo `AGENTS.md`	model-facing version test	10 pairs; 8 routes
Executable policy	Gatekeeper and Trivy	exact policy differences	16 cases; 4 edges
Boundary faults	Cosign, Agent Governance, agentgateway	verifier and adapter validity	6, 2, 33 cases
Synthetic structure	SLSA-inspired migration	rule and edge coverage	60 pairs; min. 10
Evidence state	imbalanced, balanced, factor-complete	action and route diagnostics	150, 90, 15 pairs

4.2 Route studies and auxiliary instruments

The model-facing study sends the ten `AGENTS.md` pairs to eight hosted routes for three repetitions, totaling 480 scheduled calls. Every request includes the complete pinned policy file. Primary endpoints use the first repetition: coverage, item and pair accuracy, changed-pair action invariance, stable-control accuracy, action recall, and the full confusion matrix. Repeated results measure within-route stability.

The matched factor diagnostic selects the lexicographically first balanced-challenge pair for each of 15 registered factors. Seven routes receive the same 30 items for three repetitions, totaling 630 scheduled calls. Selection uses factor identifiers and precedes every route output. Each cell permits one request and zero client retries. The two historical balanced-challenge routes are recomputed on the same subset without new calls.

Prior instruments provide depth and controls. A Gatekeeper study repeats 24 items three times on two routes sharing the same provider, base URL, API, prompt-only mode, and adapter settings. A SLSA-inspired migration defines four actions over structured attestation fields and covers 12 rules and ten reachable edges (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). The deterministic executor is the *definitional policy ceiling*; policy-aware and profile-blind random forests are learned controls. The evidence-state suite contains 300 imbalanced items, including eight `DEMOTE` and eight `ESCALATE` labels. Its Naive Bayes control supplies a weak shortcut check. The public action-balanced challenge contains 90 pairs and 15 factors, with 30 `HOLD`, 30 `DEMOTE`, 30 `ESCALATE`, and 90 `ALLOW` members. We fix its cases and labels before inference. Its estimand is robustness on known cases.

5 Results

5.1 Source validation and finite structural coverage

Open Policy Agent execution reproduces both Gatekeeper changes on all 12 contexts. A two-pair witness covers 2/2 changed rules and both `ALLOW`→`HOLD` and `HOLD`→`ALLOW` transitions, meeting the two-edge lower bound. Trivy reproduces two transitions in each direction around its inverted predicate, with a two-case minimum. Cosign reproduces four verifier fault modes and requires four cases to cover them.

Replaying the fixed revision's 15 upstream tests against the Agent Governance Toolkit parent yields 11 passes; all 15 pass at the change. Both decision cases change from an undefined deny verdict to `deny`, yielding two mapped `ALLOW`→`HOLD` transitions and a one-case minimum. For agentgateway, we inspect the parent source and execute tests at the change commit, where all 33 selected tests pass. Together these revisions instantiate policy, verifier, and guardrail-interface faults.

The external policy receipt verifies both `AGENTS.md` parents and changes, source bytes, licenses, and maintainer reviews. The ten-case mapping contains six changed pairs and four stable controls: 14 `ALLOW`, three `HOLD`, two `ESCALATE`, and one `DEMOTE` item per replication. These are source-validation and author-mapping claims; route behavior is reported separately.

The four-action migration gives a separate structural result. Its 12-rule stratified suite covers 12/12 changed rules and 6/10 reachable edges. The graph suite covers 12/12 rules and 10/10 edges. An exact search selects a 10-pair, 20-call joint suite, reducing the 120-call graph suite by 83.3%. Mistral and

Phi-4 score 10/60 and 2/60 pairs on the full graph. This minimum certifies coverage of the declared rules and observed edges. Stochastic fault detection and release sufficiency remain unmeasured. The registered policy-aware random-forest control reaches 107/120 items and 47/60 pairs, while the profile-blind control reaches 57/120 items and 0/60 pairs. The zero-pair result shows that paired scoring detects policy-profile invariance on this graph. The weak Naive Bayes control reaches 184/300 items and 34/150 pairs on the imbalanced suite, with balanced action recall 0.328.

5.2 MODEL-FACING POLICY REVISIONS

Table 2 reports the designated first replication on the two public `AGENTS.md` revisions. Across eight routes, 132/160 scheduled items produce valid decisions, 128/160 actions are correct, and 121/160 exact decisions match both action and registered reason codes. Of 80 pairs, 55 are complete and 52 are correct, giving 52/80 end-to-end and 52/55 conditional pair accuracy.

Table 2: External policy revisions, first replication. Changed and stable columns retain their 6- and 4-pair denominators. Conditional item/pair scores use valid items or complete pairs.

Route	Valid	E2E item	E2E pair	Changed	Stable	Cond. item/pair
GLM 5.3	16/20	16/20	6/10	2/6	4/4	16/16; 6/6
Luna 5.6	16/20	16/20	6/10	3/6	3/4	16/16; 6/6
HY3	18/20	18/20	8/10	4/6	4/4	18/18; 8/8
Kimi K2.7 Code	18/20	18/20	8/10	5/6	3/4	18/18; 8/8
MiMo 2.5 Pro	19/20	18/20	9/10	5/6	4/4	18/19; 9/9
MiniMax 2.7	16/20	15/20	5/10	1/6	4/4	15/16; 5/6
OX Alpha	12/20	12/20	4/10	1/6	3/4	12/12; 4/4
Qwen 3.7 Plus	17/20	15/20	6/10	2/6	4/4	15/17; 6/8

The changed-policy subset contributes 23/48 correct pairs end to end and 23/26 conditional on both decisions. Three of the 26 complete changed pairs return the same action across source versions. Stable controls contribute 29/32 correct pairs end to end and 29/29 conditional, so successful routes usually preserve the action when the mapped policy outcome is unchanged. Six routes recover the single `DEMOTE` item; OX Alpha and Qwen Plus miss it. Across all three repetitions, 387/480 items and 158/240 pairs are correct end to end, compared with 387/399 items and 158/169 complete pairs conditionally. The 81 terminal failures comprise 26 server-side transport failures, nine length-terminated JSON failures, and 46 outputs whose action conflicts with their reason codes. Decision counts include valid outputs only; the end-to-end denominator remains 480.

5.3 MATCHED FACTOR-COMPLETE ROUTES

Table 3 reports the predesignated first replication. Decision coverage spans 3/30 to 29/30, end-to-end item accuracy spans 3/30 to 26/30, and pair accuracy spans 0/15 to 11/15. The two historical anchors, executed earlier under different route states, reach 23/30 items and 8/15 pairs for Qwen Max and 14/30 items and 2/15 pairs for Flash. They are temporal anchors from an earlier service state.

Table 3: Factor-complete first replication. Conditional scores use valid items or complete pairs; all other columns retain the full 30-item or 15-pair denominator.

Route	Valid	E2E item	E2E pair	Conditional item / pair
GLM 5.3	9/30	9/30	1/15	9/9; 1/1
Luna 5.6	29/30	26/30	11/15	26/29; 11/14
HY3	3/30	3/30	0/15	3/3; –
Kimi K2.7 Code	16/30	15/30	4/15	15/16; 4/5
MiMo 2.5 Pro	16/30	11/30	1/15	11/16; 1/5
MiniMax 2.7	9/30	7/30	0/15	7/9; 0/2
Qwen 3.7 Plus	27/30	25/30	10/15	25/27; 10/12

Across three repetitions, Luna produces 87/90 valid decisions, 79/90 correct items, and 34/45 correct pairs; Qwen Plus produces 80/90, 69/90, and 26/45. Their all-three correct-pair counts are 8/15 and

5/15. The remaining routes complete at most 48/90 decisions and at most 6/45 pairs. Of 311 terminal failures, 194 are JSON syntax failures, 87 schema failures, eight semantic failures, and 22 transport failures. A length or token limit accompanies 280/281 syntax and schema failures. This concentration attributes the dominant loss to the tested route contract and output ceiling. Conditional columns describe only the surviving subset; the end-to-end columns define route performance. Among the five DEMOTE items, Luna recovers 3/5 and Qwen Plus 2/5; the other five routes recover 0/5. Appendix B gives every action recall and confusion cell.

5.4 EXECUTABLE-POLICY ROUTE REPETITIONS

On the repeated Gatekeeper test, Flash reaches 53/72 raw-strict items and 21/36 pairs; Qwen reaches 10/72 items and 0/36 pairs. All 63 extracted actions match the executable oracle. Interface losses come from 62 Qwen fenced objects and, for Flash, 12 read timeouts, three fenced objects, three non-JSON envelopes, and one length termination. Parsing is repair-free. Across repetitions, 13/24 Flash items and 5/12 pairs pass every time; Qwen has zero fully stable items. Appendix B reports each repetition and exact-decision score.

5.5 ENDPOINT SENSITIVITY AND EVIDENCE-STATE DIAGNOSTICS

The endpoint reanalysis covers 1,320 previously scheduled items across ten route stacks. Raw-strict parsing finds 69 extractable actions outside the full six-field contract; 49 actions and 40 pairs are correct. Conversely, 452 contract-valid registered-parser actions are unavailable to raw-strict parsing, including 329 correct actions and 78 correct pairs. Most of the latter occur on historical local routes whose registered parsers used deterministic single-object extraction. Neither endpoint contains the other, and each answers a different integration question.

On the SLSA Qwen route, raw-strict action yields 120/120 items and 60/60 pairs, while the full contract yields 88/120 and 28/60. On the balanced Qwen challenge, the registered parser accepts six fenced objects, including four correct actions and two correct pairs, that raw-strict parsing rejects. Raw-strict scores are therefore 128/180 items and 46/90 pairs, compared with 132/180 and 48/90 under the full contract. The balanced Flash route changes from 105/180 and 19/90 under raw-strict parsing to 90/180 and 13/90 under the contract. Registered-parser actions and full-contract actions coincide in these records because a parser-owned decision is retained only after contract validation.

Table 4: Auxiliary item/pair metrics. All raw and contract scores are end to end; the last column is conditional on contract validity. "–" marks unavailable retained raw text.

Route	Reg. valid	Raw item/pair	Contract item/pair	Cond. item/pair
Imbalanced Qwen	272/300	–	258/300; 113/150	258/272; 113/123
Challenge Qwen	148/180	128/180; 46/90	132/180; 48/90	132/148; 48/64
Challenge Flash	146/180	105/180; 19/90	90/180; 13/90	90/146; 13/61
SLSA Qwen	88/120	120/120; 60/60	88/120; 28/60	88/88; 28/28
SLSA Flash	108/120	110/120; 51/60	108/120; 49/60	108/108; 49/49

Table 5: End-to-end contract-action recall by gold action.

Route	ALLOW	HOLD	DEMOTE	ESCALATE
Imbalanced Qwen	143/150	110/134	0/8	5/8
Challenge Qwen	79/90	26/30	4/30	23/30
Challenge Flash	68/90	9/30	0/30	13/30

Action support changes the diagnosis. Table 5 gives end-to-end contract-action recall. The imbalanced suite has eight examples for each rare action; Qwen recovers 0/8 DEMOTE and 5/8 ESCALATE. The balanced challenge increases each non-ALLOW action to 30 cases: Qwen recovers 4/30 DEMOTE, and Flash recovers 0/30. The complete confusion matrices in Appendix B retain terminal failures in every row.

6 DISCUSSION

Policy revision becomes a regression coordinate. The two `AGENTS.md` histories supply independently authored policy changes that can alter the prescribed action for a fixed scenario. Binding the source version, scenario, declared action, and route record creates a test that can be replayed after either the policy or service stack changes. Among complete first-replication pairs, 23/26 changed pairs follow the declared transition and 29/29 stable controls preserve the declared action. The three invariant changed pairs show the diagnostic value of pairing even when item accuracy is high. Executable policies add exact rule and transition certificates. Prose policies retain source traceability while leaving the action mapping with the study author or policy owner.

A route has two failure surfaces. The model must select the prescribed action and complete the service contract. All 63 extracted Gatekeeper actions agree with the executable oracle, while transport and serialization failures remove many complete pairs. In the factor diagnostic, output limits accompany 280/281 syntax and schema failures. The external study adds a third interface fault: 46 outputs pair an action with inconsistent reason codes. Conditional accuracy isolates action choice after interface completion; end-to-end accuracy measures the delivered route. Parser, contract, output ceiling, and denominator therefore belong in a deployment-facing test specification.

Pairs and action balance expose complementary shortcuts. The profile-blind control completes 0/60 SLSA-inspired pairs because it ignores the policy profile. The imbalanced evidence-state suite assigns only eight items to each rare action and obscures weak `DEMOTE` behavior behind frequent `ALLOW` and `HOLD` cases. The balanced challenge and matched factor study make that failure visible. Pair accuracy tests joint version sensitivity, action recall tests label-specific behavior, and the terminal column measures interface completion.

Prospective use. A policy owner can register changed and stable cases before merging a source revision, run them against candidate service routes, and retain the frozen records as a release receipt. Fully structured inputs should use a deterministic executor, which defines the policy ceiling in this study. Prose policies require an explicit mapping and independent review appropriate to the deployment. The Naive Bayes result serves only as a weak shortcut check. Deployment utility remains an empirical question for the adopting system.

7 LIMITATIONS

The external study contains two policy revisions and ten author-constructed pairs. Upstream maintainers authored the source text; we authored the scenarios, precedence rule, action mapping, and labels. Human annotation, deployment telemetry, and prevalence estimation remain outside this study. It measures frozen-case conformance to the declared mapping. A field study with policy-owner labels and observed downstream consequences would establish operational validity.

The remaining histories form a convenience sample of policy, verifier, and adapter faults. The fixed source screen covered 465 repository seeds and inspected 11 candidates. Synthetic instruments provide controlled action and factor support; the distribution of repository policies and user requests remains unmeasured.

Every structural certificate is relative to a finite context set and registered witnesses. Missing contexts can hide behavior, and incorrect witnesses invalidate rule coverage. Minimum-suite optimality covers the declared rule and edge universe; stochastic fault-detection power and release sufficiency remain unmeasured. Trivy uses a documented compatibility helper for historical Rego; Cosign isolates verifier logic under a fixed policy.

Hosted results identify route configurations, which combine a provider-defined model identifier, prompt, API adapter, output constraint, service state, and time. Serving internals are opaque, and most breadth routes share one provider. The comparisons support route-level descriptions; model-family rankings and causal ablations remain unsupported. The factor-complete historical anchors ran at a different time and serve only as temporal references. OX Alpha is a provider-designated stealth route: its underlying model identity and data-retention and training terms were undisclosed at freeze time.

The historical Flash lane for the imbalanced suite used an obsolete service interface: all 300 requests returned HTTP 401 before model output. Later Flash routes establish that the newer service paths execute. The simultaneous changes in service and output stack preclude a causal comparison. The balanced challenge was frozen before inference and estimates robustness on known public cases. Policy retrieval, ambiguous-source adjudication, evidence extraction, and downstream utility remain outside the evaluation.

8 CONCLUSION

CLAIMPROMOTEBENCH treats policy version as a first-class regression input. It binds source revisions to paired cases, prescribed action transitions, parser provenance, complete denominators, and, for finite executable policies, exact structural certificates. Public `AGENTS.md` revisions extend the protocol to independently authored model-facing policy text. Their paired cases expose both version-sensitive behavior and three invariant changed-pair observations. Controlled instruments reveal rare-action and interface failures that item accuracy alone can conceal. The artifact supports auditable pre-merge and pre-deployment checks with a narrow estimand: route conformance on frozen cases. A deterministic executor remains the definitional policy ceiling for structured inputs. Field studies with policy-owner labels and downstream outcomes are the next step toward deployment claims.

9 REPRODUCIBILITY STATEMENT

The released artifact binds policies, datasets, prompts, source tests, route configurations, and parser implementations. Its deterministic replay recomputes source bindings, coverage, minimum suites, endpoint metrics, action recalls, and confusion matrices from the existing hosted records. The anonymous bundle removes private service metadata and hidden reasoning.

10 AI ASSISTANCE DISCLOSURE

Generative AI assisted literature search, code development, analysis checks, and editing. The author selected the research question and protocol, verified sources, results, and artifacts, and assumes responsibility for the paper.

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A Enumerator and Reducer Guarantees

Minimum-suite hardness. Take a Set Cover instance with universe U and subsets $S_1, \ldots, S_n$. Create one required rule for each element of U and one candidate context for each S_i whose rule-witness set is S_i. Assign every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover U. A minimum joint suite therefore gives a minimum set cover.

Exact dynamic program. Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first j candidates, retain for every reachable mask the shortest subset, breaking ties by context identifier. Excluding candidate $j + 1$ preserves each state; including it unions its coverage mask with each prior mask. Induction on j proves that the final full-mask representative is cardinality-minimum. The implementation matches exhaustive enumeration on 64 deterministic small candidate pools.

Edge lower bound. Each pair realizes one directed edge. Covering k edges requires at least k pairs. If the specialized search selects one candidate per edge and also covers every rule, its k-pair witness meets the lower bound. This certifies the two-pair Gatekeeper suite and the ten-pair SLSA-inspired suite relative to their declared candidate pools.

B Complete Action-Level Results

Table 6: Gatekeeper route results by repetition. All columns are end to end. R is raw-strict action; F is exact full decision. Contract action equals R in every row.

Route	Repetition	R item	F item	R pair	F pair
Qwen	1	3/24	3/24	0/12	0/12
Qwen	2	2/24	2/24	0/12	0/12
Qwen	3	5/24	5/24	0/12	0/12
	total	10/72	10/72	0/36	0/36
DeepSeek Flash	1	15/24	15/24	6/12	6/12
DeepSeek Flash	2	19/24	18/24	7/12	6/12
DeepSeek Flash	3	19/24	19/24	8/12	8/12
	total	53/72	52/72	21/36	20/36

Table 7: External policy-revision first-replication confusion matrices. `N` is a terminal route failure; every row retains its full gold support.

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
GLM	ALLOW	12	0	0	0	2
GLM	HOLD	0	1	0	0	2
GLM	DEMOTE	0	0	1	0	0
GLM	ESCALATE	0	0	0	2	0
Luna	ALLOW	12	0	0	0	2
Luna	HOLD	0	1	0	0	2
Luna	DEMOTE	0	0	1	0	0
Luna	ESCALATE	0	0	0	2	0
HY3	ALLOW	13	0	0	0	1
HY3	HOLD	0	2	0	0	1
HY3	DEMOTE	0	0	1	0	0
HY3	ESCALATE	0	0	0	2	0
Kimi	ALLOW	13	0	0	0	1
Kimi	HOLD	0	2	0	0	1
Kimi	DEMOTE	0	0	1	0	0
Kimi	ESCALATE	0	0	0	2	0

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
MiMo	ALLOW	13	0	0	1	0
MiMo	HOLD	0	2	0	0	1
MiMo	DEMOTE	0	0	1	0	0
MiMo	ESCALATE	0	0	0	2	0
MiniMax	ALLOW	11	0	0	1	2
MiniMax	HOLD	0	1	0	0	2
MiniMax	DEMOTE	0	0	1	0	0
MiniMax	ESCALATE	0	0	0	2	0
OX	ALLOW	10	0	0	0	4
OX	HOLD	0	1	0	0	2
OX	DEMOTE	0	0	0	0	1
OX	ESCALATE	0	0	0	1	1
Qwen+	ALLOW	11	1	0	1	1
Qwen+	HOLD	0	2	0	0	1
Qwen+	DEMOTE	0	0	0	0	1
Qwen+	ESCALATE	0	0	0	2	0

Table 8: End-to-end action recall for every route summarized in the main text. "–" marks actions outside a task's label set.

Configuration and endpoint	ALLOW	HOLD	DEMOTE	ESCALATE
Gatekeeper Qwen, raw-strict	4/36	6/36	–	–
Gatekeeper Flash, raw-strict	32/36	21/36	–	–
Imbalanced Qwen, contract	143/150	110/134	0/8	5/8
Challenge Qwen, raw-strict	77/90	26/30	3/30	22/30
Challenge Qwen, contract	79/90	26/30	4/30	23/30
Challenge Flash, raw-strict	68/90	10/30	2/30	25/30
Challenge Flash, contract	68/90	9/30	0/30	13/30
External GLM 5.3, rep. 1	12/14	1/3	1/1	2/2
External Luna 5.6, rep. 1	12/14	1/3	1/1	2/2
External HY3, rep. 1	13/14	2/3	1/1	2/2
External Kimi K2.7, rep. 1	13/14	2/3	1/1	2/2
External MiMo 2.5, rep. 1	13/14	2/3	1/1	2/2
External MiniMax 2.7, rep. 1	11/14	1/3	1/1	2/2
External OX Alpha, rep. 1	10/14	1/3	0/1	1/2
External Qwen 3.7+, rep. 1	11/14	2/3	0/1	2/2
Factor GLM 5.3, rep. 1	7/15	1/5	0/5	1/5
Factor Luna 5.6, rep. 1	14/15	4/5	3/5	5/5
Factor HY3, rep. 1	3/15	0/5	0/5	0/5
Factor Kimi K2.7, rep. 1	10/15	3/5	0/5	2/5
Factor MiMo 2.5, rep. 1	9/15	0/5	0/5	2/5
Factor MiniMax 2.7, rep. 1	6/15	0/5	0/5	1/5
Factor Qwen 3.7+, rep. 1	14/15	5/5	2/5	4/5
SLSA Qwen, raw-strict	60/60	20/20	20/20	20/20
SLSA Qwen, contract	60/60	20/20	1/20	7/20
SLSA Flash, raw-strict	59/60	17/20	17/20	17/20
SLSA Flash, contract	59/60	17/20	15/20	17/20
Graph Mistral, contract	9/36	18/24	0/24	13/36
Graph Phi-4, contract	21/36	5/24	0/24	1/36

Table 9: Factor-complete first-replication confusion matrices. N is a terminal route failure. Every row retains its full gold support.

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
GLM	ALLOW	7	0	0	0	8
GLM	HOLD	0	1	0	0	4
GLM	DEMOTE	0	0	0	0	5
GLM	ESCALATE	0	0	0	1	4
Luna	ALLOW	14	1	0	0	0
Luna	HOLD	0	4	0	0	1
Luna	DEMOTE	2	0	3	0	0
Luna	ESCALATE	0	0	0	5	0
HY3	ALLOW	3	0	0	0	12
HY3	HOLD	0	0	0	0	5
HY3	DEMOTE	0	0	0	0	5
HY3	ESCALATE	0	0	0	0	5
Kimi	ALLOW	10	0	0	0	5
Kimi	HOLD	0	3	0	0	2
Kimi	DEMOTE	1	0	0	0	4
Kimi	ESCALATE	0	0	0	2	3
MiMo	ALLOW	9	0	0	2	4
MiMo	HOLD	0	0	0	0	5
MiMo	DEMOTE	3	0	0	0	2
MiMo	ESCALATE	0	0	0	2	3
MiniMax	ALLOW	6	0	0	0	9
MiniMax	HOLD	0	0	0	0	5
MiniMax	DEMOTE	2	0	0	0	3
MiniMax	ESCALATE	0	0	0	1	4
Qwen+	ALLOW	14	0	0	1	0
Qwen+	HOLD	0	5	0	0	0
Qwen+	DEMOTE	1	0	2	0	2
Qwen+	ESCALATE	0	0	0	4	1

Table 10: Gatekeeper aggregate raw-strict confusion counts over three repetitions (left) and SLSA-inspired transition-graph contract-action confusion counts (right). N marks the endpoint's terminal failure category.

Gatekeeper

Route	Gold	ALLOW	HOLD	N
Qwen	ALLOW	4	0	32
Qwen	HOLD	0	6	30
Flash	ALLOW	32	0	4
Flash	HOLD	0	21	15

SLSA-inspired graph

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Mistral	ALLOW	9	8	0	9	10
Mistral	HOLD	2	18	0	4	0
Mistral	DEMOTE	3	0	0	0	21
Mistral	ESCALATE	2	3	0	13	18
Phi-4	ALLOW	21	0	0	0	15
Phi-4	HOLD	1	5	0	0	18
Phi-4	DEMOTE	3	12	0	0	9
Phi-4	ESCALATE	0	0	0	1	35

Table 11: Imbalanced-suite and balanced-challenge full-contract confusion counts. Columns are terminal predicted actions; N is a contract failure.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Imbalanced Qwen	ALLOW	143	0	0	0	7
Imbalanced Qwen	HOLD	4	110	0	2	18
Imbalanced Qwen	DEMOTE	8	0	0	0	0
Imbalanced Qwen	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
Challenge Qwen	HOLD	0	26	0	0	4
Challenge Qwen	DEMOTE	8	4	4	4	10
Challenge Qwen	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
Challenge Flash	HOLD	13	9	0	0	8
Challenge Flash	DEMOTE	25	3	0	0	2
Challenge Flash	ESCALATE	0	3	0	13	14

Table 12: Balanced-challenge raw-strict action confusion counts. N marks a terminal extraction failure.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Challenge Qwen	ALLOW	77	0	0	0	13
Challenge Qwen	HOLD	0	26	1	0	3
Challenge Qwen	DEMOTE	7	3	3	4	13
Challenge Qwen	ESCALATE	0	0	0	22	8
Challenge Flash	ALLOW	68	4	5	13	0
Challenge Flash	HOLD	13	10	7	0	0
Challenge Flash	DEMOTE	25	3	2	0	0
Challenge Flash	ESCALATE	0	3	2	25	0

Table 13: SLSA-inspired raw-strict and contract-action confusion counts. N is the endpoint's terminal failure category.

Route	Endpoint	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Qwen	raw-strict	ALLOW	60	0	0	0	0
Qwen	raw-strict	HOLD	0	20	0	0	0
Qwen	raw-strict	DEMOTE	0	0	20	0	0
Qwen	raw-strict	ESCALATE	0	0	0	20	0
Qwen	contract	ALLOW	60	0	0	0	0
Qwen	contract	HOLD	0	20	0	0	0
Qwen	contract	DEMOTE	0	0	1	0	19
Qwen	contract	ESCALATE	0	0	0	7	13
Flash	raw-strict	ALLOW	59	0	0	0	1
Flash	raw-strict	HOLD	0	17	0	0	3
Flash	raw-strict	DEMOTE	0	0	17	0	3
Flash	raw-strict	ESCALATE	0	0	0	17	3
Flash	contract	ALLOW	59	0	0	0	1
Flash	contract	HOLD	0	17	0	0	3
Flash	contract	DEMOTE	0	0	15	0	5
Flash	contract	ESCALATE	0	0	0	17	3